

Building a Cloud-Agnostic Test Environment **with Terraform**

Terraform is a tool to safely and predictably create, change and improve infrastructure.

Terraform is cloud-agnostic and can use a single configuration to handle multiple cloud providers.

It provides cost benefits, scalability, reliability, stability and tools for failure analysis, while enabling better collaboration and visibility.



Stable and reliable infrastructure is one of the most important factors for any test automation project as the ROI is achieved when automated suites are executed regularly in a controlled environment. Other factors are the ease of scaling the infrastructure to handle the increase in the volume of automated tests and how to reduce the total execution time.

The architecture proposed below describes one of the optimal solutions to address these challenges, leading to stability, reliability and cost benefits. This solution uses Infrastructure-as-a-Code to handle the dynamics of infrastructure creation, based on requirements at runtime on the selected configuration.

Why choose Terraform?

In the current business scenarios, it's quite challenging to choose one cloud provider when there are multiple options, each with its own set of benefits. Also, from the deployment point of view, deploying multiple clouds increases the fault-tolerance of the system.

Terraform is cloud-agnostic, and allows a single configuration to be used to manage multiple providers and to even handle cross-cloud dependencies. In the present DevOps culture, automated tests are an integral part of deployment pipelines. Therefore Terraform is an obvious choice to simplify management and orchestration, helping operators build large scale multi-cloud infrastructures.

The benefits

Listed below are some of the key benefits of using cloud infrastructure with Terraform to run automated test suites:

1. A cloud-agnostic environment with Infrastructure-as-a-Code
2. Cost benefits
3. Stability
4. Scalability
5. Reliability
6. Good quality execution videos for failure analysis
7. Better collaboration and visibility

The components of the proposed architecture design

Figure 2 shows the proposed architecture design. The components of this design are described below.